

This will connect to the server specified in the configuration file and will search from the base mentioned there.

LDAP is a TCP/IP protocol and an LDAP server binds to TCP port 389 by default. When using TLS, the default secure LDAP port is 636. Instead of specifying a host, an LDAP URI can also be used. LDAP URIs are of the form: `ldap://host:port`

The above LDAP search using LDAP URI instead of the hostname can be executed as follows:

```
[vbg@vbg-work scratch]$ ldapsearch -x -H ldap://x500.bund.de \
-b "o=bund.de"
```

...and the corresponding entry in `/etc/openldap/slapd.conf` will be:

```
URI ldap://x500.bund.de
```

Installing the OpenLDAP server

To query other directory servers is one thing and to be able to create your own directory server to store and serve data is quite another. The first step towards creating your own directory server is installing the required software. In Fedora, a *yum install openldap-servers* will do the job.

For the purpose of this article, I will create the *openldap-server* on a virtual RHEL 5.4 server and the client will reside on my virtual Fedora 12 host. The details of the test set-up are:

1. LDAP server hostname: `vbg.knaff.org` (192.168.122.1) – RHEL 5.4 (64 bit)
2. LDAP client hostname: `vbg-work.knaff.org` (192.168.122.244) – Fedora 12 (32 bit)

The LDAP server software installs the stand-alone LDAP daemon (*slapd*). The version of the OpenLDAP server on RHEL 5.4 is 2.3 (*openldap-servers-2.3.43-3.el5*). The *slapd* daemon binds on Port 389 and serves client requests based on its directory data. In Fedora/RHEL systems, the important locations are:

- `/usr/sbin/slapd` – the stand-alone LDAP daemon binary program
- `/var/lib/ldap/` – the folder containing directory data files
- `/etc/openldap/slapd.conf` – the LDAP server configuration file
- `/etc/init.d/ldap` – init script for the LDAP daemon

It also creates the user and group LDAP. For security purposes, the *slapd* daemon runs with the privileges of this user:

```
[root@vbg ~]# cat /etc/passwd | grep ldap
ldap:x:55:55:LDAP User:/var/lib/ldap:/bin/false
```

Configuring the LDAP server

The main configuration file is `/etc/openldap/slapd.conf`.

conf. For the init script to be able to read this file, the permissions of this file should be set to 640 and the owner group should be set to `ldap`:

```
[root@vbg openldap]# ls -l /etc/openldap/slapd.conf
-rw-r----- 1 root ldap 3731 Nov 6 2008 /etc/openldap/slapd.conf
```

Let's start configuring our very own LDAP server. As always, remember to keep a back-up of the main config file—just in case something goes wrong:

```
[root@vbg openldap]# cp -a slapd.conf slapd.conf.org
[root@vbg openldap]# vim /etc/openldap/slapd.conf
```

The 'include' section of this config includes certain files from the schema directory. These files describe the structure of our directory. We will examine these in more detail in another article. For now, to get going quickly, the other important directives in the configuration file are:

```
database    bdb # this defines the database backend, used by openldap.
             # In this case - Berkeley Data Base Backend
suffix      "dc=my-domain,dc=com" # This is the root of the Directory Tree
rootdn      "cn=Manager,dc=my-domain,dc=com"
             # This is the Distinguished Name of the superuser of the directory
# rootpw    secret
```

For a directory server, the DN and the base suffix are important concepts.

The base suffix of a directory specifies the top of the directory tree (much like "/" signifies the root of the Linux filesystem). All directory data is stored under this base suffix, much like folders and files in the traditional Linux filesystem. The DN (Distinguished Name) is the unique name of a directory entry (much like file names and directories—which need to be unique within the same parent folder).

LDAP is generally associated with DNS as most organisations would want their domain names mapped to their directory data. Therefore, if my domain name is *knaff.org*, this domain context can be represented in the directory as follows:



Therefore, in the directory, the domain *knaff.org* can be represented in a hierarchical fashion as: `dc=knaff, dc=org`